



INVENTORY MANAGEMENT

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**A PROJECT REPORT SUBMITTED IN PARTIAL
FULFILLMENT OF THE REQUIREMENT FOR THE DEGREE
BACHELOR OF ENGINEERING IN COMPUTER ENGINEERING**

**FACULTY OF ENGINEERING & INTERNATIONAL COLLEGE
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Computer Engineering Project
entitled
INVENTORY MANAGEMENT

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คำสำคัญ : ลาเท็ก / วิทยานิพนธ์ (~5 คำ)

13 หน้า

INVENTORY MANAGEMENT

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ABSTRACT

This thesis documents the design, development, and evaluation of a three-tier Enterprise Resource Planning (ERP) system prototype specifically engineered to meet the operational and financial control needs of small to medium-sized enterprises (SMEs). The project addresses the prevalent issue of fragmented data and inefficient workflows in growing businesses by integrating core functions—Partner Management, Product & Stock Control, Purchasing, and Sales—into a singular, cohesive platform.

The primary objective was to construct a robust, real-time application using a standard web stack to ensure accessibility and maintainability. This involved leveraging Structured Query Language (SQL) for the persistent and transactional data layer, housing all master and ledger information. The presentation layer utilizes HTML and CSS to deliver a responsive, user-friendly interface for all data entry and viewing requirements, connected by a server-side application layer responsible for executing all complex business logic.

The methodology focused on modeling and implementing critical ERP logic, including a weighted moving average algorithm for accurate product costing and a sophisticated Document & Calculation Logic module designed to handle multi-level discounts and various VAT configurations. A central achievement was the successful implementation of transactional posting logic, which ensures real-time synchronization of inventory levels (Stock IN/OUT) and financial entries (Accounts Payable/Receivable)

immediately upon invoice posting. Furthermore, the system incorporates essential validation checks, such as automated credit limit verification and stock availability checks before authorizing sales transactions.

The key finding of the evaluation is the validation of the system's integrated architecture, which successfully provided a unified, real-time operational dashboard and eliminated data integrity issues common in non-integrated systems. The Reporting & Analytics module demonstrated the capability to accurately calculate profit margins and generate vital performance reports directly from the SQL-stored transactional history.

In conclusion, this project successfully delivers a comprehensive, web-based ERP solution that leverages relational database technology (SQL) to establish data integrity and provides significant benefits in process automation, inventory accuracy, and managerial decision-making capacity for scaling businesses.

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CHAPTER 1

INTRODUCTION

1.1 Background

Inventory management is a critical function for small and family-run businesses, particularly for middle-man companies that must track purchases, sales, and product stock across multiple suppliers. However, many existing inventory management systems are complex, feature-heavy, and designed for professional corporate users. As a result, elderly business owners or non-technical staff often struggle to use these systems effectively. This challenge is also experienced in my own family business, where existing software tools prove difficult to operate despite repeated training. These limitations highlight the need for a simpler, more intuitive system that better aligns with the abilities and workflow of small middle-man operations.

Current inventory systems such as EazyStock, Zoho Inventory, and QuickBooks provide stock tracking and document management features, but they often require monthly subscription fees and lack user-friendly interfaces for older users. Additionally, most available solutions do not integrate the full range of documents and operational processes that middle-man businesses rely on, such as purchase requisitions, purchase orders, delivery orders, tax invoices, receipts, and credit or debit notes. These businesses also require real-time visibility of stock, clear tracking of process statuses, and the ability to calculate essential inventory values such as reorder points, safety stock, and optimal order quantity.

Furthermore, although some commercial systems include basic chatbot functions, they cannot retrieve detailed operational data or answer business-specific questions. There is a clear gap for an AI-assisted interface that allows users—especially elderly or non-technical staff—to ask questions in natural language and receive information directly from the system’s backend data. Such a conversational tool can simplify access to stock information, document status, process tracking, and inventory calculations without requiring navigation through complex menus.

Therefore, this project aims to develop a user-friendly inventory management system tailored to the workflow of middle-man businesses. The system records purchase and sales transactions, tracks process status, manages and links essential business documents, displays warehouse stock in real time, and supports the calculation of key inventory formulas. In addition, the project integrates an AI chatbot using the ChatGPT API to allow users to request specific information from the system through natural language queries. Developing this system provides a cost-effective, easy-to-use solution that improves daily operations while reducing reliance on expensive commercial software.

1.2 Objective

1. To develop a system that records purchase and sales transactions, including the current status of each process, with an attachment file function that links all supporting documents directly to the corresponding transactions.
2. To develop an inventory management system that displays the overall product stock in the warehouse and provides tools for calculating essential inventory
3. To integrate an AI chatbot using the ChatGPT API that connects to the system's backend data, allowing users to request specific information through natural language queries.

1.3 Scope

1. The scope of this project includes the development of a system capable of recording purchase and sales transactions and tracking the status of all operational processes. This includes stages such as quotation, pending approval, in-transit delivery, quality checking, completed processes, and cancelled transactions. The system provides a file-upload mechanism that allows users to attach and link supporting documents directly to each transaction. All essential business documents—such as Purchase Requisitions (PR), Purchase Orders (PO), Delivery Orders (DO), Goods Received Notes (GRN), invoices, receipts, and credit or debit notes—are supported and stored systematically within the database to ensure accurate

reference and operational transparency.

2. The inventory management component of the system displays the available stock within the warehouse and provides real-time visibility into stock movement and overall inventory status. The system incorporates several calculation tools to support inventory planning and decision-making. These include the Reorder Point (ROP), Safety Stock, Economic Order Quantity (EOQ), Order-Up-To Level under the P-System, and Z-Score for service-level determination. By computing these values numerically and presenting them to the user, the system assists in managing inventory efficiently and maintaining optimal stock levels.
3. The project also integrates an AI chatbot designed to connect directly to the database and retrieve specific information for calculation and operational decision support. Users interact with the chatbot through text-based natural language queries, enabling them to extract precise information from the system based on their needs. The chatbot does not process images or voice messages; instead, it focuses solely on text commands to access relevant data such as stock quantities, document information, and transaction details. This provides a simplified, user-friendly method for accessing system information without navigating complex interfaces.
4. Limitations: Restrict project validation to the use of simulated transactional data, excluding high-volume concurrent user testing or live production environment performance metrics. Limit the audit trail functionality solely to the Activity Log of document status changes.
5. Ranges of Data, Date, and Time: Support an unlimited range of historical data in the design; however, validate reporting using a simulated 12-month transactional history. All critical stock, credit, and costing calculations operate on a real-time basis following any document posting.
6. Platform(s)/Device(s) Used in This Project: The project utilizes a Three-Tier Web Architecture. The data layer employs MySQL as the Relational Database Management System (RDBMS), which serves as the reliable platform for storing and managing structured data relationships. The backend logic is handled by Django, while the client interface is built with React.js. The system is optimized primarily

for desktop web browsers to support complex data entry and reporting workflows effectively.

1.4 Expected Results

1. The expected result of this project is the development of a functional inventory management system that can record purchase and sales transactions, track the status of each process, and link all supporting documents directly to their corresponding transactions. The system is expected to provide an up-to-date view of stock levels in the warehouse after user approval of each transaction, together with clear traceability of documents and process history. In addition, the system should be able to calculate key inventory values—such as Reorder Point (ROP), Safety Stock, Economic Order Quantity (EOQ), Order-Up-To Level under the P-System, and Z-Score—based on correctly structured and consistent data stored in the database.
2. The project also aims to enhance user accessibility by integrating an AI chatbot that allows elderly or non-technical users to retrieve information from the system through natural language text queries. The chatbot is expected to return accurate data from the database, using the right parameters needed for the inventory calculation formulas defined in the system. Overall, the completed system is expected to make daily operations easier, reduce manual checking, and support better-informed decisions without relying on complex commercial software.

1.5 Timeline

CHAPTER 2

LITERATURE REVIEW

This chapter reviews three key works that directly support the development of this project. The first reference, A Full-Stack Web Solution for Online FMCG Management System Using MERN, informs the design of the web-based platform, guiding the system architecture, interface structure, and implementation approach used for recording transactions and managing documents. The second reference, Essentials of Inventory Management, provides the theoretical foundation for structuring the purchase, sales, and stock management workflow while also supplying the core formulas applied in this project, including Reorder Point, Safety Stock, EOQ, the P-System, and Z-Score calculations. The third reference, Aisha: A Custom AI Library Chatbot Using the ChatGPT API, contributes the methodological framework for integrating an AI chatbot capable of retrieving and summarizing system data through natural language queries. Together, these works provide the technological, operational, and AI-driven basis for building a complete inventory management system with document linkage, structured calculations, and a chatbot interface designed to improve accessibility for elderly or non-technical users.

2.1 A Full-Stack Web Solution for Online FMCG Management System Using MERN: Design, Implementation, and Evaluation [1]

This project presents the comprehensive development of a Full-Stack Web Solution for an Online Fast-Moving Consumer Goods (FMCG) Management System. Designed to modernize and streamline retail operations for FMCG products, the system leverages the power of the MERN stack (MongoDB, Express.js, React, and Node.js) for high availability, scalability, and data integrity. The article details the three core phases of the project: Design, Implementation, and Evaluation. The design phase focuses on developing an intuitive, responsive, and secure architecture, including multi-user role support (e.g., admin, staff, supplier) and key modules for inventory management, order

processing, real-time stock updates, and automated billing. The implementation phase outlines the deployment of the MERN stack components, creating a robust backend for API integration and a dynamic frontend for a seamless user experience. Finally, the evaluation phase assesses the system's effectiveness in enhancing operational efficiency, reducing manual effort, and bridging the gap between traditional retail methods and modern digital solutions, catering to the needs of retailers, wholesalers, and distributors.

2.2 Essentials of Inventory Management [2]

Timeless stock-keeping fundamentals meet up-to-the minute technologies to optimize efficiency and drive profits! Inventory management is about more than counting what you've got. It's about understanding business realities and making decisions that balance current demand with future needs—while keeping overhead and operating costs to a minimum. Now in its Second Edition, *Essentials of Inventory Management* gives inventory professionals the information they need to maximize productivity in key areas, from physical stock issues to problem identification and resolution to technologies like RFID and other automated inventory mechanisms. Perfect for novice and veteran managers alike, this ultra-practical book covers topics such as: Forecasting and replenishment strategies • Differences between retail and manufacturing inventories • Materials requirements planning and just-in-time inventory systems • Simple formulas for calculating quantities and schedules • Management of inventory as a physical reality and a monetary value • Supply chain risk management Complete with detailed examples, handy tools, and a revised and expanded chapter analyzing “Why Inventory Systems Fail and How to Fix Them,” this nontechnical yet thorough guide is perfect for both instructional and on-the-job use.

2.3 Aisha: A Custom AI Library Chatbot Using the ChatGPT API [4]

This article describes the development of “Aisha,” a custom chatbot for a university library, built using Python and the ChatGPT API. Aisha is designed to provide reference and support services (e.g., answering library queries) outside normal operating hours. The paper discusses the chatbot's architecture, capabilities, limitations, and the

development process. It positions this as one of the early attempts to deploy a ChatGPT-based system for library services and provides insight into customization, prompt design, integration, and user feedback.

CHAPTER 3

METHODOLOGY

Describe what you did so that others can follow to recreate your work/system/ experiment proposed in this project. You should provide addiquet details about your work, system, algorithm, or design. This includes technical methods used (e.g., machine learning techniques, library, tools), data collection and preparation/pre-Processing, hyperparameter configurations, measurements, evaluation approaches or techniques used to evaluate your work performance. (It is recommended to writing this chapter after completing CH2 Literature Review)

3.1 System Design

You may start with an overview of your system/algorithm design by using the fol- lowings:

- Flowchart
- System, Design, or Block Diagram
- Phedu Code
- Others (figures or even tables)

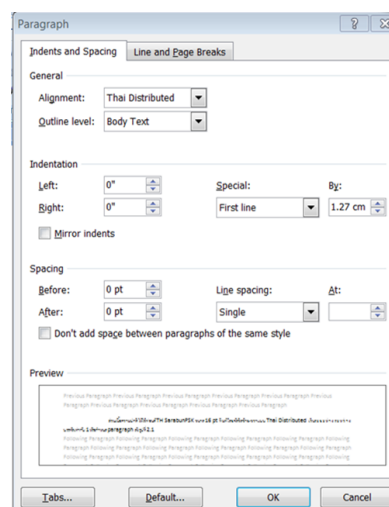


Figure 3.1 Paragraph arrangement (example)

3.2 Data

3.3 Methods

3.4 Hyperparameter Configurations

3.5 Performace Evaluation

CHAPTER 4

RESULT

This chapter reports your experiment results or study report proposed in previous chapter. It typically consists of 1) Results in terms of tables or figures and 2) explanation or discussion of your results, study, or key findings.

4.1 Results

(Experiment/Results in the forms of tables or/and figures)

4.2 Discussions

(Interpretation or the meaning of your experiment/results)

CHAPTER 5

CONCLUSION

This chapter summarizes all of your work. It typically starts with the brief explanation of what you do such as the objective of your work, design, data, experiment results, key findings, interpretation of your experiment and/or result. The obstacles of your work can also be discussed and finally followed by future work.

5.1 Conclusion

...

5.2 Obstacles

...

5.3 Future Work

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REFERENCES

[1] .

[2] .

[3] .

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